School of Computer Science Engineering and Technology

Course- **BTech**
Course Code- **CSET340**

Year- **2026**
Date- **13-04-2026 to 17-04-2026**

Type- **Specialization Elective**
Course Name-**Advanced Computer vision and Video Analytics**
Semester- **EVEN**
Batch- **2023-2027 (V Sem)**

Lab Assignment No. 09-10

Exp. No.	Name	CO-1	CO-2	CO-3
1.	To perform camera calibration and extract various projection matrix.	✓	✓	✓

Objective: Apply camera calibration methods and generate the projection matrix and other intrinsic and extrinsic parameters. via the use of Python using OpenCV/Pillow/Ski Images Library?

Perform the task provided as follows.

Task 1: Linear Camera Calibration (DLT Method) [4 Marks]

Use the provided 3D world coordinates of the checkerboard corners and their corresponding 2D image coordinates from a single view to estimate the **Projection Matrix P.**

Task 2: Distortion Analysis & Correction [4 Marks]

Use a set of checkerboard images to perform full camera calibration including non-linear parameters.

Task 3: Performance Metrics [2 Mark]

Calculate the **Reprojection Error** to quantify the accuracy of your calibration.

Note:- **Suggested Platform:** Python: Jupyter Notebook/Visual Studio Code/Google Colab.
Mode of Delivery: Face-to-face: Instructor-led discussion and live coding demonstration. Hands-on Practice via Google Colab/VS code/Notebook.
Submission: On LMS within the prescribed time frame, else marks will.